

TE Curve Verification Pipeline Directional Model Comparison

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report is the reduced selected-model TE Curve Verification Pipeline offline comparison for one dataset and one direction. It includes only the active selected model-development candidates plus reduced baseline anchors.

Dataset And Split

- dataset config: `config/datasets/transmission_error_dataset.yaml` ;
- dataset root: `data\simplified_dataset` ;
- comparison mode: `reduced_selected_model_matrix` ;
- candidate count: 9 ;
- held-out curve count before candidate filtering: 97 ;
- percentage-error denominator: `peak_to_peak_truth` ;
- this reduced report excludes `global` candidates by design;
- only candidates valid for this direction are evaluated.

Candidate Inventory

Candidate	Family	Source	Kind	Surface	Valid Direction
polished_rcim_model_bank_reproduction_GBM19_Bw	GBM	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward
polished_rcim_model_bank_reproduction_ET19_Bw	ET	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward
polished_feedforward_Bw	feedforward	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_tree_Bw	tree	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_harmonic_regression_Bw	harmonic_regression	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_periodic_gru_sequence_Bw	periodic_gru_sequence	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_periodic_mlp_harmonic_Bw	periodic_mlp_harmonic	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave4_3_mixture_density_k3_Bw	wave4_3_mixture_density_k3	polished_model_development_registry	wave1_registry_model	Bw	backward
wave52b_offset_centered_shape_harmonic_Bw	wave52b_offset_harmonic_guided	wave52b_offset_harmonic_guided_registry	wave1_registry_model	Bw	backward

Backward Comparison

Wave 5.2B Offset And Harmonic Guided Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave52b_offset_centered_shape_harmonic_Bw	0.020690	0.021139	48.162	112.613

Polished Model-Development Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_harmonic_regression_Bw	0.003678	0.004012	8.058	15.071
polished_wave4_3_mixture_density_k3_Bw	0.005049	0.005359	11.212	23.958
polished_periodic_gru_sequence_Bw	0.006320	0.006795	14.078	28.125
polished_feedforward_Bw	0.008079	0.008672	17.892	35.715
polished_periodic_mlp_harmonic_Bw	0.009393	0.009831	20.198	50.399
polished_tree_Bw	0.015215	0.015523	34.609	61.506

Polished RCIM Model-Bank Reproduction Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_rcim_model_bank_reproduction_ET19_Bw	0.007426	0.007704	16.960	62.118
polished_rcim_model_bank_reproduction_GBM19_Bw	0.007624	0.007918	17.434	48.611

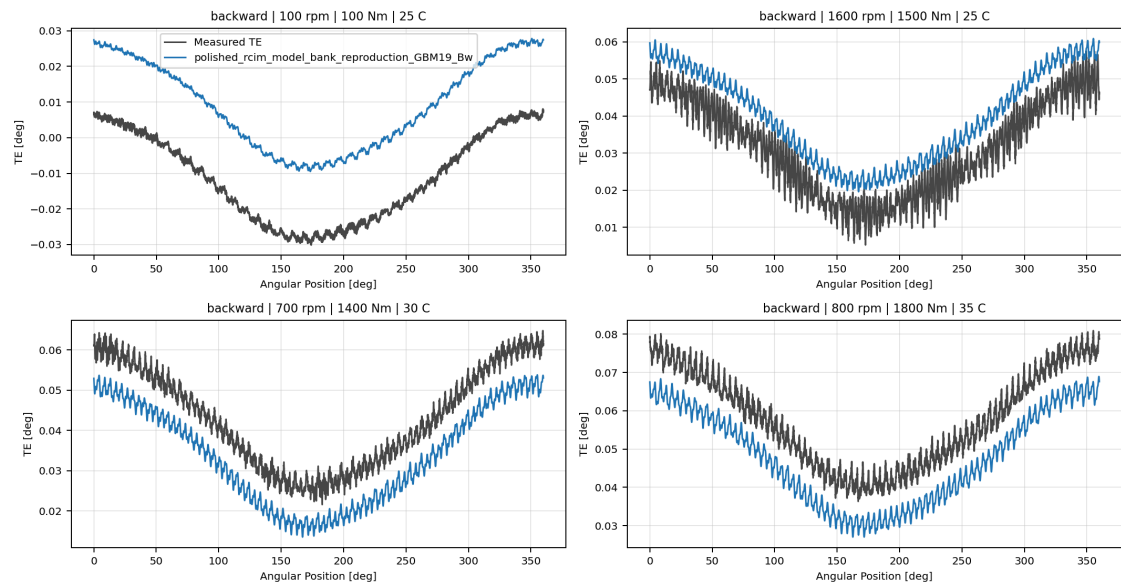
Curve Evidence

Each selected candidate is shown with four deterministic held-out curves. The dark line is the measured TE curve and the blue line is the model prediction for the same operating condition.

Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
polished_rcim_model_bank_reproduction_GBM19_Bw	Selected candidate	0.007624	0.007918	17.434	48.611
polished_rcim_model_bank_reproduction_ET19_Bw	Selected candidate	0.007426	0.007704	16.960	62.118
polished_feedforward_Bw	Selected candidate	0.008079	0.008672	17.892	35.715
polished_tree_Bw	Selected candidate	0.015215	0.015523	34.609	61.506
polished_harmonic_regression_Bw	Surface leader	0.003678	0.004012	8.058	15.071
polished_periodic_gru_sequence_Bw	Selected candidate	0.006320	0.006795	14.078	28.125
polished_periodic_mlp_harmonic_Bw	Selected candidate	0.009393	0.009831	20.198	50.399
polished_wave4_3_mixture_density_k3_Bw	Selected candidate	0.005049	0.005359	11.212	23.958
wave52b_offset_centered_shape_harmonic_Bw	Selected candidate	0.020690	0.021139	48.162	112.613

polished_rcim_model_bank_reproduction_GBM19_Bw

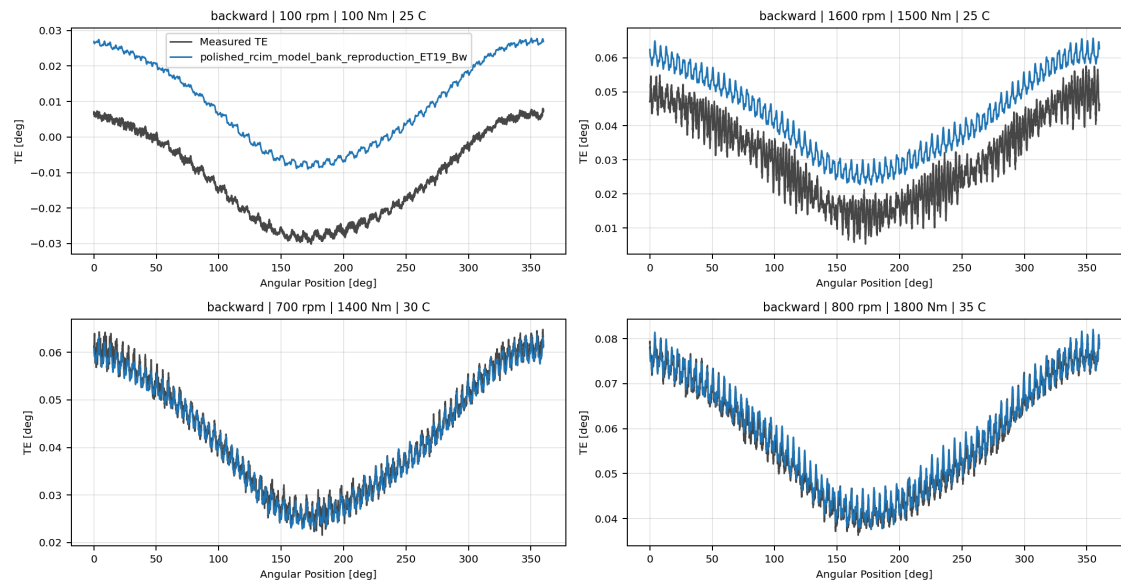
polished_rcim_model_bank_reproduction_GBM19_Bw



polished_rcim_model_bank_reproduction_GBM19_Bw measured-versus-predicted TE collage

polished_rcim_model_bank_reproduction_ET19_Bw

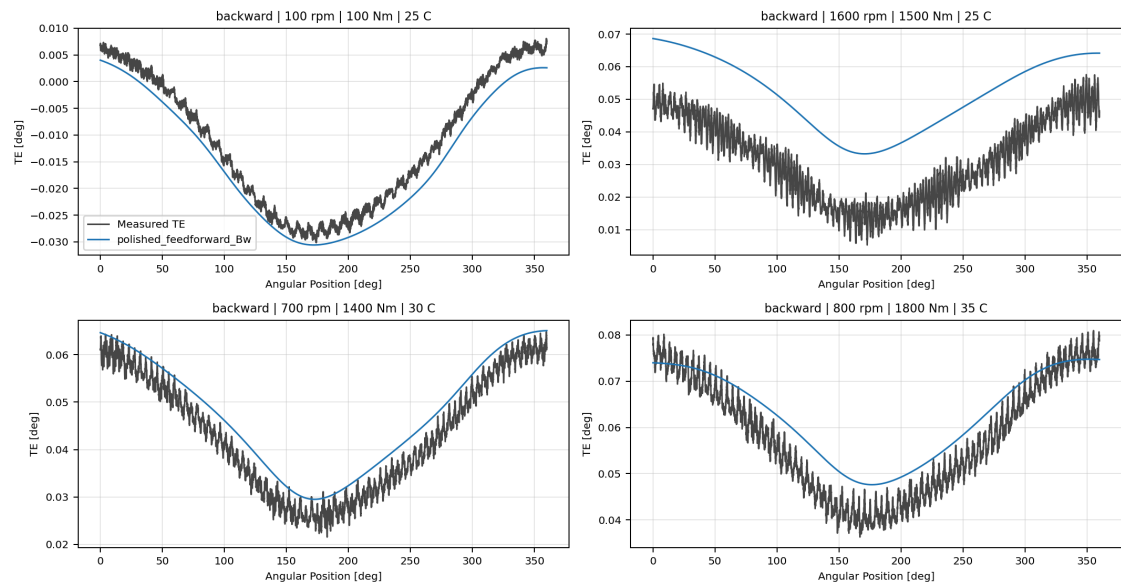
polished_rcim_model_bank_reproduction_ET19_Bw



polished_rcim_model_bank_reproduction_ET19_Bw measured-versus-predicted TE collage

polished_feedforward_Bw

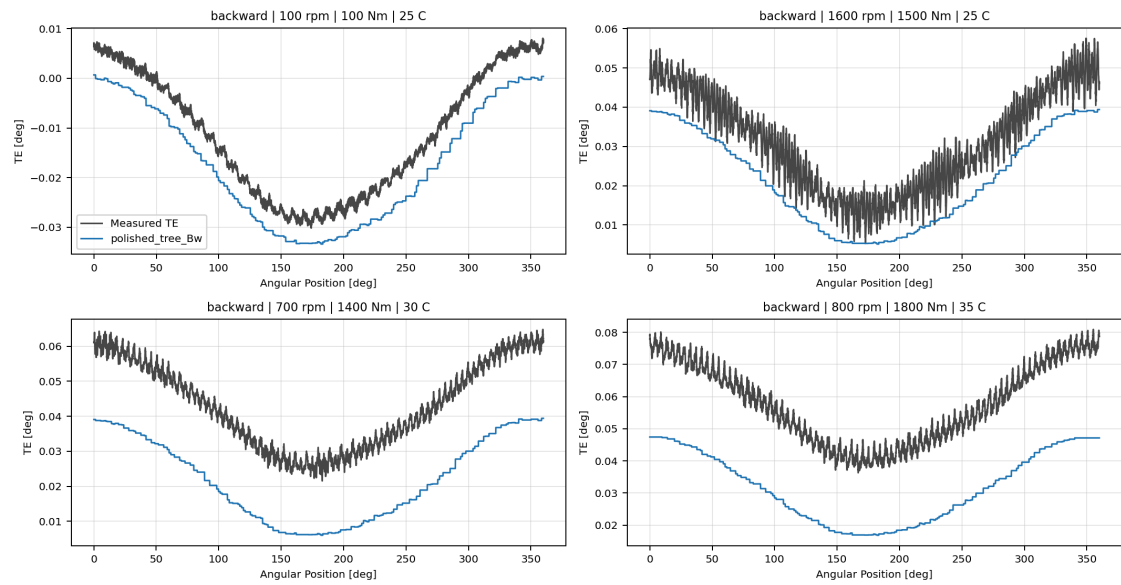
polished_feedforward_Bw



polished_feedforward_Bw measured-versus-predicted TE collage

polished_tree_Bw

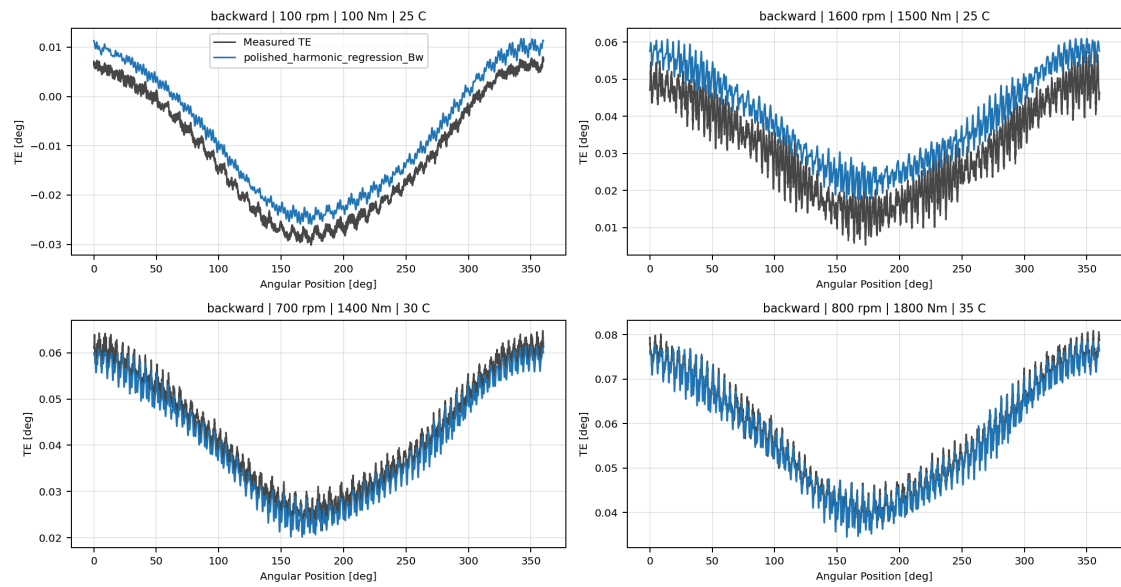
polished_tree_Bw



polished_tree_Bw measured-versus-predicted TE collage

polished_harmonic_regression_Bw

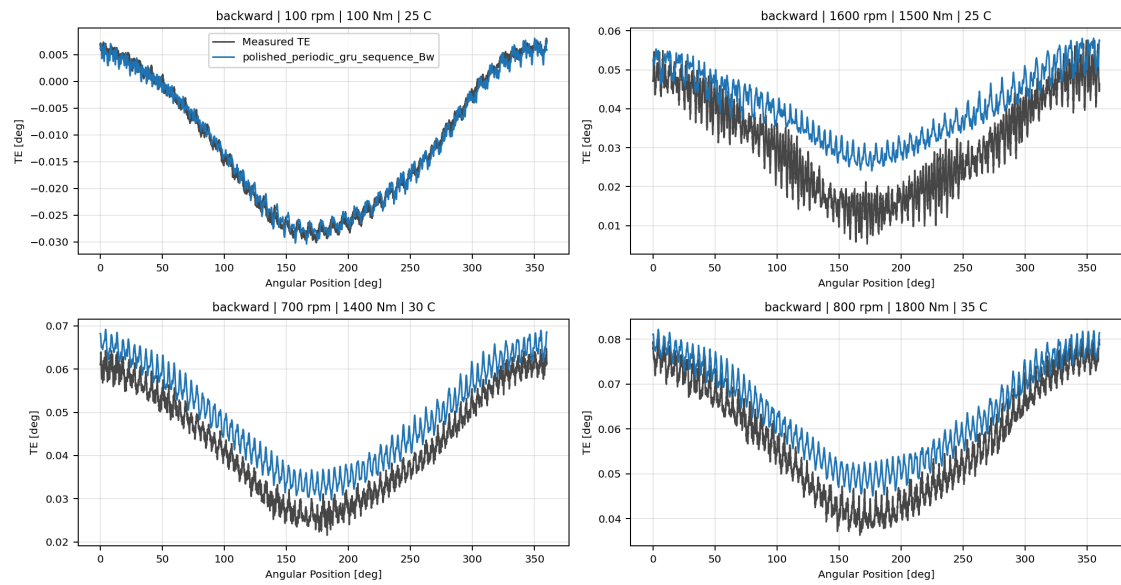
polished_harmonic_regression_Bw



polished_harmonic_regression_Bw measured-versus-predicted TE collage

polished_periodic_gru_sequence_Bw

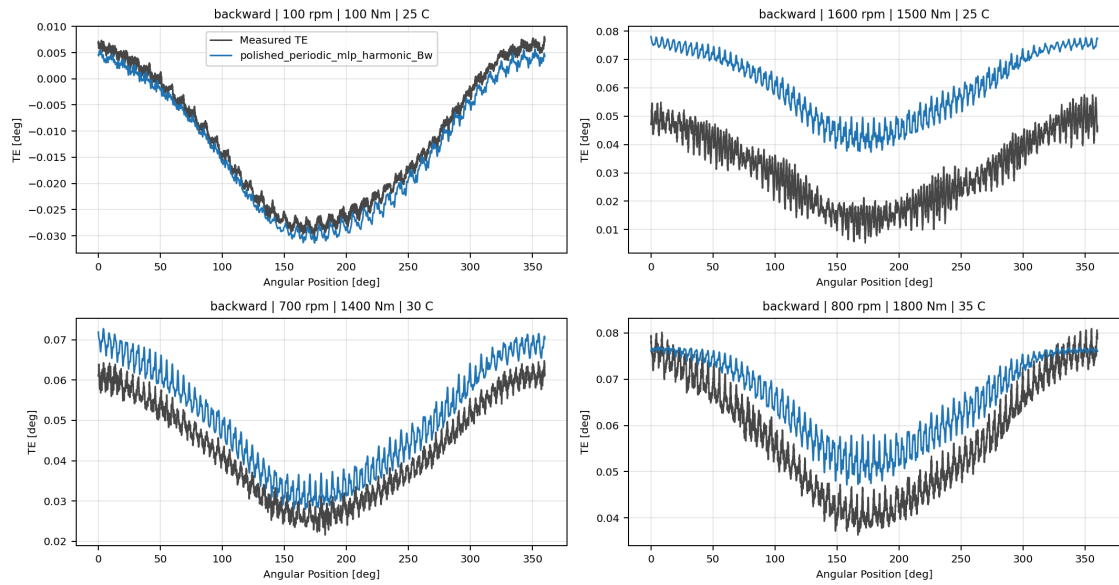
polished_periodic_gru_sequence_Bw



polished_periodic_gru_sequence_Bw measured-versus-predicted TE collage

polished_periodic_mlp_harmonic_Bw

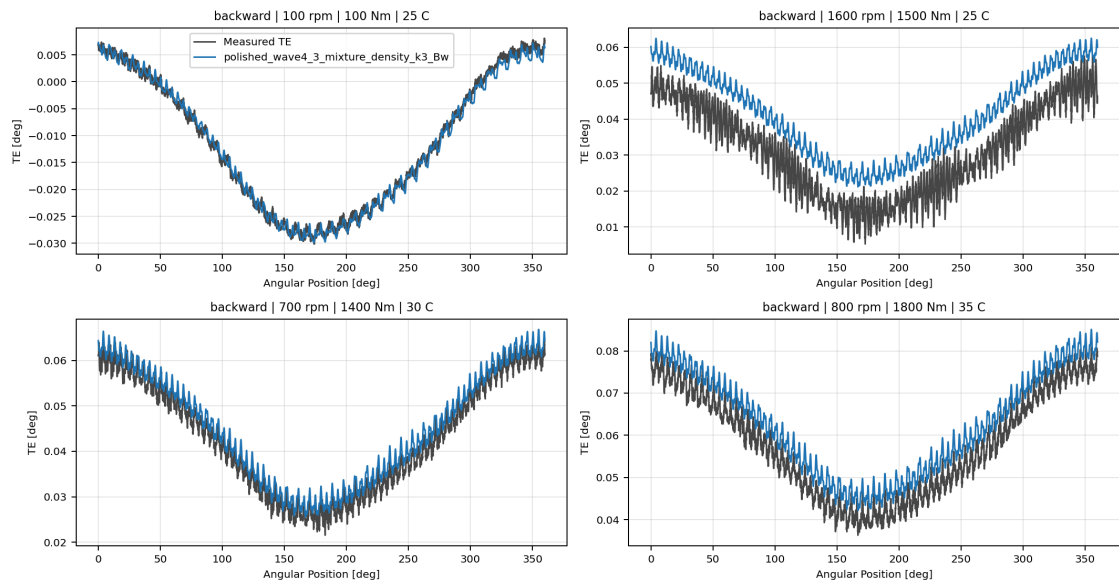
polished_periodic_mlp_harmonic_Bw



polished_periodic_mlp_harmonic_Bw measured-versus-predicted TE collage

polished_wave4_3_mixture_density_k3_Bw

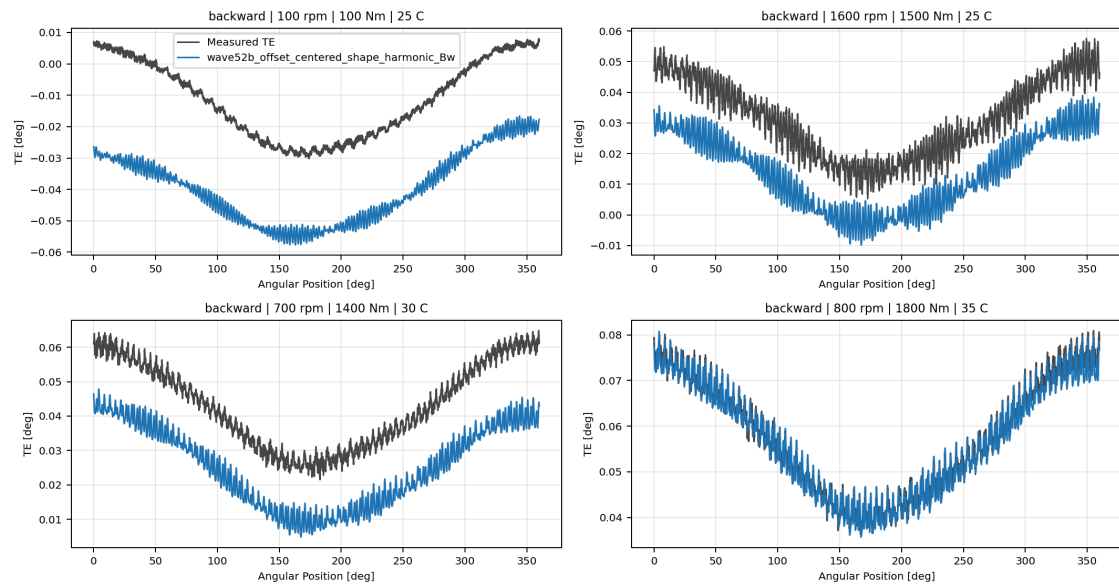
polished_wave4_3_mixture_density_k3_Bw



polished_wave4_3_mixture_density_k3_Bw measured-versus-predicted TE collage

wave52b_offset_centered_shape_harmonic_Bw

wave52b_offset_centered_shape_harmonic_Bw



wave52b_offset_centered_shape_harmonic_Bw measured-versus-predicted TE collage

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-06-19-55-29_track2_reduced_selected_model_matrix_track2_selected_simplified_dataset_backward_2026_07_06\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-06-19-55-29_track2_reduced_selected_model_matrix_track2_selected_simplified_dataset_backward_2026_07_06\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots\paused_reduced_selected_model_matrix` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within each source group and direction. Directional candidates are never evaluated on the opposite direction in this reduced report. `global` is paused for the active reduced pipeline and is not included in the candidate inventory. The `rcim_track1` forward reference banks use the opposite stored `h0` sign convention relative to the TE Curve Verification Pipeline reconstruction contract, so the TE Curve Verification Pipeline comparison applies the documented source-specific `h0` compatibility multiplier before curve reconstruction.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online `Table 9` compensation benchmark.
- The report uses the saved Python model artifacts from `models/` ; ONNX parity checks remain a separate deployment-readiness task.